Weekly Report: Ultra-High-Pressure ODMR — Should we introduce a lock-in amplifier?

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Question

> A lock-in cannot beat shot noise, it can only move the measurement above the technical $1/f$ noise, so at what detector noise knee f_{knee} does modulated detection actually start to improve the precision σ_D of the fitted ODMR line centre, and which modulation scheme should we use?

Hypothesis

If the technical noise has relative PSD $S(f) = S_0(1 + f_{knee}/f)$ with S_0 pinned to the shot-noise floor, then a DC sweep — which reads the line out at $\sim 1/T_{sweep} \sim 2\,\mathrm{Hz}$ — should degrade as $\sqrt{f_{knee}}$ while modulated schemes stay flat, so there should be a break-even f_{knee} below which plain DC averaging is better because modulation throws away duty cycle.

Evidence / Interpretation

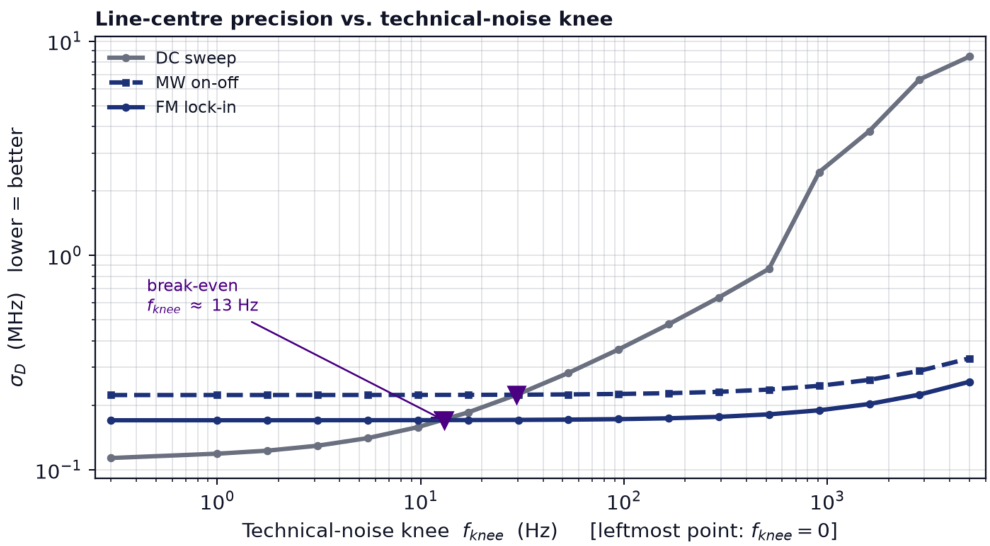

図 1: Monte-Carlo σ_D (200 realisations, identical total measurement time and identical noise realisation per trial) for a DC sweep, MW on-off chopping and FM lock-in, as a function of the technical-noise knee. Triangles mark the break-even knees.

- The break-even is at $f_{knee} \approx 13\,\text{Hz}$ for FM and $\approx 30\,\text{Hz}$ for on-off chopping: below that, DC averaging is genuinely better (0.113 vs 0.170 MHz at $f_{knee} = 0$, the cost of the modulation duty cycle), and above it DC collapses while both modulated schemes stay flat to within 35% out to 5 kHz.
- At a realistic $f_{knee} = 1\,\text{kHz}$ the gain is $13.7\times$ in σ_D, FM beats on-off by $\approx 1.3\times$ at its optimal dither depth of $0.5\,\Delta\nu$, and the gain saturates once f_{mod} exceeds f_{knee} — so a few kHz of modulation is enough and nothing is gained by going faster.

Gap / Failure

- The simulation assumes purely multiplicative intensity noise, so it misses additive drifts (background creep, detector dark-current wander) that modulation rejects even better — the quoted gains are therefore a lower bound.
- Our own f_{knee} is unmeasured, and since the whole decision turns on that single number the answer is currently a criterion rather than a recommendation.

Next Step

Digitise a few seconds of the MW-off photodetector record and take its power spectral density to read off f_{knee} directly, which both settles the decision and shows whether software demodulation of that same record already delivers the gain without buying hardware.

References

1. A. Dréau *et al.*, "Avoiding power broadening in optically detected magnetic resonance of single NV defects," Phys. Rev. B **84**, 195204 (2011).
2. J. F. Barry *et al.*, "Sensitivity optimization for NV-diamond magnetometry," Rev. Mod. Phys. **92**, 015004 (2020).
3. Repository: `code/lockin_sim.py`, `code/fig7_lockin.py` (this analysis).